



Introduction to Electronic Engineering / 电子工程导论



Opening 开场

Good afternoon everyone!

My name is **[Daniel]**, and before I start, let me share something interesting.

Many of my friends think I'm a **computer science student** because I've built a lot of software projects — like a *cinema broadcasting system*, an *airport announcement generator*, my *personal website*, and even an *online aviation chart viewer*.

But actually, my real major is **Electronic Engineering** — and that's exactly what I'll be talking about today.



What is Electronic Engineering 什么是电子工程?

English:

So, what exactly is electronic engineering?

In short, it's the study of **how to design, build, and improve electronic systems** — from small circuits to huge communication networks.

We learn how electric signals move, how to process them, and how to make devices smarter, faster, and more efficient.



Applications 1 — PCB & Basic Electronics 应用一：PCB 与基础电子学

English:

Everything starts from the **PCB**, or printed circuit board.

You can think of it as the **skeleton** of any electronic device — from smartphones to industrial robots.

When electronic components are mounted on it, it becomes **PCBA** — printed circuit board assembly.

Without PCBs, modern electronics simply wouldn't exist.



Applications 2 — Communication Systems 应用二：通信系统

English:

One of the most fascinating areas in electronic engineering is **communication systems**. Wi-Fi, 5G, Bluetooth, GPS, satellites — all of them rely on electronics to send and receive signals.

Even pilots depend on radio and radar systems built by electronic engineers.

These systems are based on the same principles we study — signal transmission, modulation, and filtering.

Applications 3 — Advanced Technologies 应用三：前沿科技

English:

Electronic engineering also drives cutting-edge technologies like **autonomous driving** and **artificial intelligence**.

For example, in self-driving cars, sensors collect visual and radar data, and electronic circuits process that information to make driving decisions.

Even large AI models, like chatbots or image generators, rely on **powerful electronic hardware** — chips, processors, and servers — designed by electronic engineers.

My Project Example 我的项目实例

English:

Speaking of real projects — I actually made a small **runway approach light** model!

It's a decorative desk lamp that looks like an airport runway.

But it's not just for show — it connects to the internet, reads **real-time METAR weather data** through an API, and changes color according to the actual visibility and weather at a chosen airport.

As both a flight enthusiast and an electronics student, this was a really fun way to combine my hobby and my major.

Core Theories and Real-Life Examples 核心原理与生活实例

English:

Of course, everything we build starts with **basic theories**.

For example, a **capacitor** lets alternating current (AC) pass but blocks direct current (DC) — we call it “AC through, DC block.”

An **inductor** does the opposite — “DC through, AC block.”

When we combine them, we can create **low-pass filters** and **high-pass filters**, which help us remove unwanted noise.

In real life, for example, **RC low-pass filters** are widely used in DC-DC power supplies — like phone chargers or car adapters — to smooth the voltage output and reduce high-frequency noise.

So, even something as simple as your charger uses this principle!

Conclusion 总结

English:

To wrap up — electronic engineering is a field that connects **theory and reality**.

It supports everything from communication and automation to artificial intelligence and renewable energy.

It's the invisible power behind the modern world.

So next time you use your phone, send a message, or charge your laptop — remember, **electronic engineering is quietly making it all possible**.

Thank you for listening!